Bernard Lappay

blappay1721@gmail.com | San Diego, CA | blappay1721.github.io | linkedin.com/in/blappay1721 | github.com/blappay1721

WORK EXPERIENCE

Silent Studio **Jun. 2026 – Present**

Technical Contributor, Mineshoku Tensei *Remote*

- Implement gameplay and backend features for a live multiplayer game using domain-specific scripting languages.
- Build race-free multi-stage boss fights using timing-based state machines with ordered failsafes.
- Keep a multi-server deployment stable through platform updates by resolving JVM dependency conflicts and documenting regressions.
- Cut a core spatial-query system's geometry data by **~86%** by identifying it as the performance bottleneck and writing a Python geometry simplifier.
- Closed a TOCTOU race-condition exploit across dozens of config files by scripting a patch rollout that verified every change with a diff.
- Migrated **40+** in-game entities from temporary spawn/despawn to persistent, state-driven designs, testing each before rollout.
- Audited **2,295** default permissions empirically by building a Python automation harness with checkpointing and pause/resume.

PROJECTS

Adaptive Math-Reasoning Inference Pipeline *Python, vLLM, LoRA, RunPod H100s | Deep Learning course, Spr. 2026*

- Placed in the **top 28%** (#29 of 105) on the class Kaggle leaderboard as a solo entrant against teams of up to 4 (68/100 final score) by improving a fixed 4B model (Qwen3-4B-Thinking-2507) at math through inference-time design instead of retraining.
- Raised multiple-choice accuracy from **73.6% to 81.6%** by building verifier-guided majority voting (3 candidates, 2-of-3 agreement) with adaptive retries that escalate the token budget and re-prompt with verifier feedback.
- Added **7 points** to the final score (61 to 68) by auditing 7,500 distilled examples, training LoRA SFT/DPO adapters, and deploying them only as a fallback on unresolved multiple-choice questions.

Persona Chatbot & Discord Bot Platform *Python, PyTorch, Hugging Face, llama.cpp, discord.js*

- Built a 16,581-conversation training set from **330,000** chat messages by removing personal info, merging duplicate user accounts, and splitting the history into conversations.
- Trained one 1.7B-parameter model to imitate **36 personas** by fine-tuning Qwen3-1.7B with QLoRA on a single 12 GB GPU (64,000 examples), adding a DPO stage deployed only if it beat the baseline on held-out chats; runs locally at 4-bit.
- Kept each persona's voice and memory consistent by building a two-index RAG pipeline (own lines for voice, past chats for facts).

Handwritten Digit Recognition *Python, NumPy, JavaScript*

- Reached **95.4%** test accuracy, classifying 1,000 MNIST digits in ~0.25 s, by building a vectorized 1-NN classifier from scratch over 7,500 training images, **~14**× faster than scikit-learn's tree-based search with identical predictions.
- Made the model explorable in any browser by porting the search to JavaScript: draw a digit and see its five nearest training images.

Self-Hosted Game Server Infrastructure *Personal Homelab*

- Migrated a live Minecraft network serving **30+** users (proxy + 3 game servers) into Docker in one day with **zero data loss** by writing a runbook and a backup-first cutover script with rollback.
- Freed **~7.5 GB** of a 16 GB host's RAM for a co-hosted LLM by starting the two largest worlds on demand and stopping them when empty; protected data with hourly, restore-tested restic backups to an append-only second machine.
- Exposed **zero** admin ports by putting the web panel behind Cloudflare Tunnel with 2FA and moving SSH and dashboards to Tailscale.
- Closed a Docker firewall bypass and ended recurring SSD dropouts by tracing them to a drive power-management firmware bug.

EDUCATION

University of California, San Diego **Expected Jun. 2027**

B.S., Computer Engineering *La Jolla, CA*

- **Relevant Coursework:** Machine Learning: Learning Algorithms; Deep Learning; Probabilistic Reasoning in AI; Computer Design and Architecture; Data Networks; Data Structures

TECHNICAL SKILLS

- **Languages:** Python, Java, C, C++, JavaScript, SystemVerilog, Verilog, MIPS Assembly, MATLAB, Bash
- **ML & AI:** PyTorch; vLLM; Hugging Face (TRL, PEFT); LoRA/QLoRA fine-tuning (SFT, DPO); llama.cpp; Ollama; NumPy
- **Linux & DevOps:** Ubuntu Server; Docker; systemd; LVM; restic; Tailscale; Cloudflare Tunnel; ufw; fail2ban; iptables; tcpdump; SSH
- **Hardware & Tools:** ModelSim; Quartus; self-checking testbenches; Git; Maven; Paper API; Vim; LaTeX